ASEKO ASIN AQUA Home

Local Home Assistant Integration via HACS

Installation, Setup and Operating Guide

For the custom Home Assistant integration "ASEKO ASIN AQUA Home"

Repository: github.com/JS-DE-Tech/hacs-aseko-asin-aqua-home-clf

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Version note

The integration is under active development. Depending on the installed release, individual labels or features may differ. This guide describes the current target state, including local TCP connectivity, optional cloud forwarding, water-level offset, container tracking and dosing-pump calibration.

Contents

1. Scope and prerequisites

This custom integration connects an ASEKO ASIN AQUA Home controller locally to Home Assistant. Data is sent from the USR-K5 gateway built into the controller to Home Assistant over TCP. MQTT and Node-RED are not required during normal operation.

Important

The integration is read-only. It evaluates the received states and measured values. Sending control commands to the pool controller is outside the scope of this integration.

Prerequisites

- Home Assistant with HACS installed.
- Network access to the local USR-K5 gateway web interface.
- TCP port 47524 available on the Home Assistant host.
- Disable any previous Node-RED TCP listener because only one application can accept the gateway connection.
- The Home Assistant server IP address must be known. The examples in this guide use 10.100.1.90.

Data flow

Local operation without cloud forwarding:

```
USR-K5 gateway  ->  Home Assistant
```

Local operation with optional cloud forwarding:

```
USR-K5 gateway  ->  Home Assistant  ->  pool.aseko.com:47524
```

2. Installation via HACS

Install the integration as a custom HACS repository.

1. Open HACS in Home Assistant.
2. Open the three-dot menu in the upper-right corner and select "Custom repositories".
3. Enter the following repository URL:

```
https://github.com/JS-DE-Tech/hacs-aseko-asin-aqua-home-clf
```

4. Choose the category "Integration" and add the repository.
5. Install "ASEKO ASIN AQUA Home" from HACS.
6. Restart Home Assistant.
7. Open "Settings -> Devices & services -> Add integration" and search for "ASEKO ASIN AQUA Home".

Update note

After a new GitHub release, HACS can offer the updated version. If the description or images are not refreshed immediately, reload the repository information in HACS and refresh the browser with Ctrl + F5.

3. Configure the USR-K5 gateway for Home Assistant

By default, the USR-K5 gateway sends its data to the ASEKO cloud endpoint. For local integration, change the target to the local Home Assistant server address.

3.1 Sign in to the gateway

8. Open the local IP address of the USR-K5 gateway in a web browser.
9. Sign in with the default credentials:

```
Username: admin
Password: admin
```

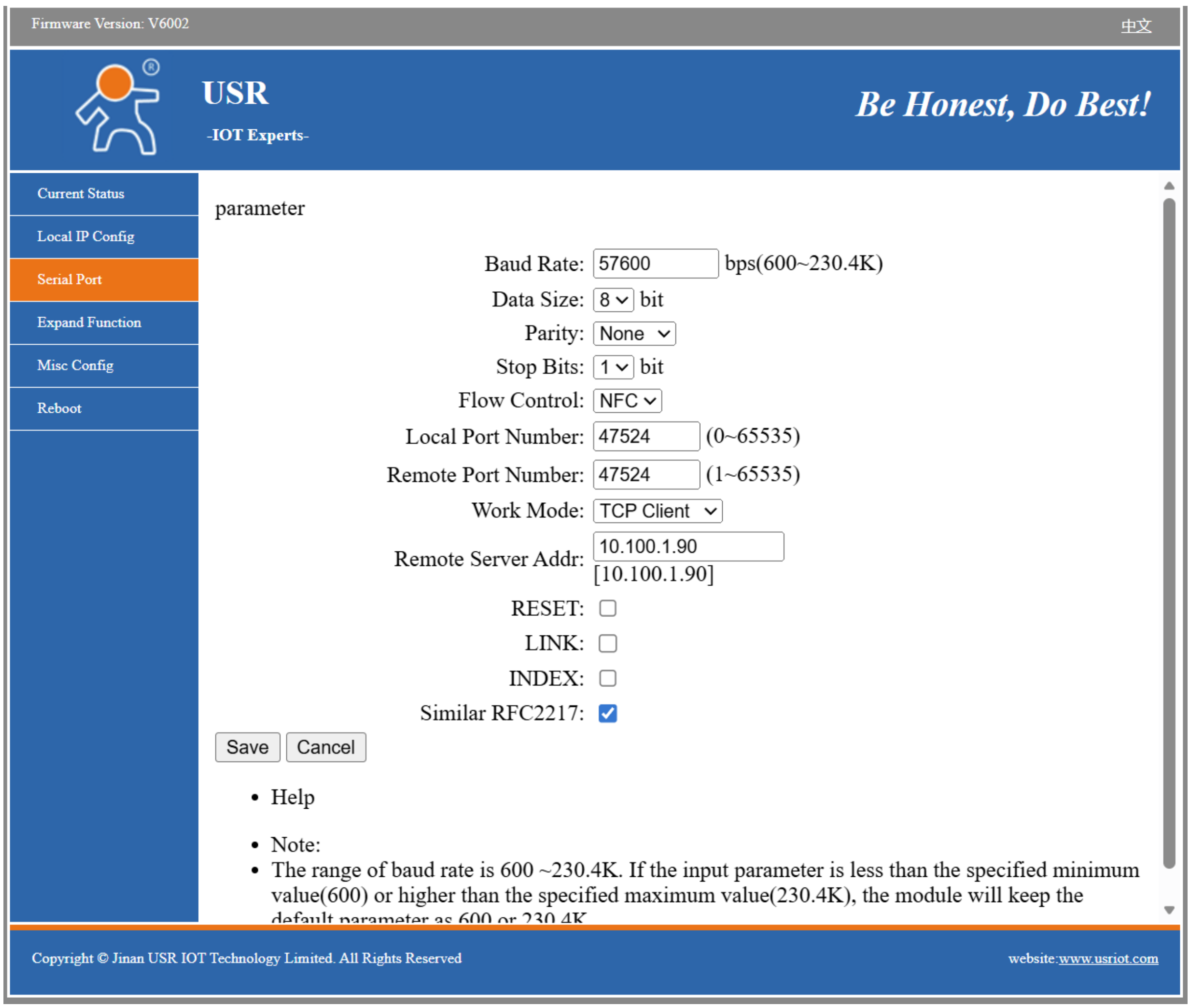
10. Open "Serial Port" in the left-hand menu.

3.2 Change the destination address

The original configuration usually points to the ASEKO cloud server:

```
Remote Server Addr:   pool.aseko.com
Remote Port Number:   47524
```

Figure 1: USR-K5 configuration in the "Serial Port" menu



Leave the serial-port settings unchanged unless you have verified that your installation uses different values:

```
Baud Rate:        57600
Data Size:        8 bit
Parity:           None
Stop Bits:        1 bit
Flow Control:     NFC
Local Port Number:  47524
Remote Port Number: 47524
Work Mode:        TCP Client
```

In the "Remote Server Addr" field, enter the local IP address or a local DNS name of your Home Assistant server. The example system uses:

```
Remote Server Addr:  10.100.1.90
Remote Port Number:  47524
```

11. Click "Save".
12. Restart the gateway if the web interface requests it or the new connection is not established immediately.
13. Add or reload the integration in Home Assistant.

Security note

The credentials admin / admin are common factory defaults. Keep the web interface accessible only from a trusted local network. Change the password after setup if the device supports it.

3.3 Alternative: local DNS redirection

As an alternative, configure a local DNS override:

```
pool.aseko.com  ->  10.100.1.90
```

This method works only when the gateway uses the hostname and receives its DNS server from the local network. DNS redirection does not apply if the gateway stores a fixed target IP address.

4. Add the integration in Home Assistant

14. Open "Settings -> Devices & services" and click "Add integration".
15. Search for "ASEKO ASIN AQUA Home".
16. Normally, keep the local listener address set to 0.0.0.0.
17. Use listener port 47524.
18. Save the integration and wait a few seconds for the USR-K5 gateway to connect.

Only one listener is possible

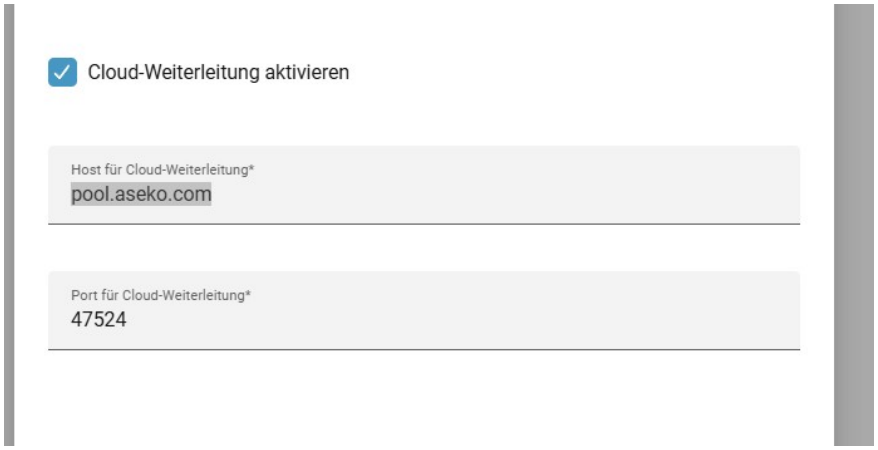
Node-RED or another program must not listen on TCP port 47524 at the same time. If the previous Node-RED flow is still active, the native integration cannot receive any data.

5. Basic settings and cloud forwarding

ASEKO cloud forwarding is optional. Local Home Assistant operation also works without a cloud connection.

5.1 Cloud forwarding

Figure 2: Options for optional cloud forwarding



Typical values:

```
Enable cloud forwarding:  Yes
Host:                     pool.aseko.com
Port:                     47524
```

Cloud forwarding sends the received original data to the ASEKO cloud endpoint in addition to the local processing path. Cloud responses are discarded and are not sent back to the gateway.

> **Failure behavior**
>
> If the cloud server is unavailable, local processing must continue to work. Older development versions may reload the complete integration after a host or port change. Newer versions should reconnect only the optional cloud forwarding session.

5.2 Water-level offset

The "Water Level Probe" sensor reports the raw probe value. The "Water Level" sensor adds the configurable offset. Example:

```
Water Level Probe:  110 cm
Offset:             33 cm
Water Level:        143 cm
```

The offset can be adjusted as a number entity in the range from -100 cm to +100 cm.

6. Entities and measured values

The integration exposes sensors, binary sensors, switches, number entities and buttons. The exact number can vary between releases.

Group	Examples	Notes
Chemistry	pH value, pH target, chlorine value, chlorine target	Chlorine values are displayed in ppm.
Temperature	Air temperature, water temperature, water-temperature target	Temperature values are displayed in °C.

Water level	Water level, probe value, low, high, refill on/off	Values are displayed in cm.
Relays	Filtration, backwash, refill, heating, chlorine, pH-minus, algicide, flocculation	Binary sensors report on/off states.
Time and maintenance	System time, deviation, backwash start, Last Backwash	Useful for status checks and maintenance records.
Pool parameters	Pool volume, backwash interval, dosing delay, maximum refill time	Pool volume is displayed in m³.

7. Container tracking and dosing-pump calibration

The controller reports the state of the dosing-pump relays. Home Assistant accumulates the runtime of each dosing pump. Once a pump flow rate is known, the integration can estimate the consumed and remaining volume.

7.1 Default container sizes

Chemical	Default size
Chlorine	20.0 l
pH-minus	20.0 l
Flocculation	6.0 l
Algicide	6.0 l

7.2 Initial setup with full containers

When the existing containers are full, press the matching "Container replaced" button once for each channel. This establishes a clean baseline for runtime tracking.

```
Pump runtime:     0 s
Consumed volume:  0.0 l
Remaining volume: full container size
Remaining level:  100.0 %
```

7.3 Calibrate the pump flow rate

Important sequence

When a container becomes empty, first press "Container empty - calculate flow rate". Only after that should you install the replacement container and press "Container replaced". Otherwise, the accumulated runtime is cleared before calibration.

19. Install a full chemical container.
20. Press the matching "Container replaced" button.
21. Leave the flow rate at 0.0 l/h until a real calibration is available.
22. Allow normal dosing operation. Home Assistant accumulates the runtime of the corresponding dosing pump.
23. When the container is actually empty, first press "Container empty - calculate flow rate".

24. Review the calculated flow rate.
25. Install a new full container.
26. Only now press "Container replaced" again.

Example calculation:

```
Container size:          20 l
Accumulated pump runtime: 18 h
Flow rate:               20 l / 18 h = 1.11 l/h
```

As long as the flow rate remains 0.0 l/h, the integration cannot provide a reliable remaining-volume estimate after the first dosing operation. In this case, remaining percentage, remaining liters and consumed liters may be displayed as unavailable. This is intentional and avoids misleading estimates.

> **Note for older releases**
>
> Older releases may show a continuously recalculated "suggested flow rate" value instead of a dedicated calibration button. This interim value becomes meaningful only after the container is actually empty. Values calculated after only a few seconds of pump runtime must be ignored.

8. Last Backwash sensor

The Last Backwash sensor records the time of the most recent successfully completed backwash cycle. A backwash is considered successful only when the backwash relay has remained active for more than 60 seconds and subsequently switches off again.

```
Backwash relay:  On
Runtime:         > 60 seconds
Backwash relay:  Off
Result:          timestamp is stored
```

Example:

```
Switched on:   03 June 2026 23:30:58
Switched off:  03 June 2026 23:32:07
Runtime:       69 seconds
Display:       03.06.2026 23:30
```

The state before the relay switches on is irrelevant. The previous state may have been off or unknown. What matters is the fully observed sequence: on for more than 60 seconds, followed by off.

9. Dashboard and practical use

The integration does not install a preconfigured dashboard. The exposed entities can be arranged in a custom Home Assistant dashboard.

Recommended dashboard areas

- Chemistry: pH value, pH target, chlorine value and chlorine target.
- Water: water temperature, temperature target, water level and thresholds.
- Operation: filtration, heating, refill and backwash.
- Dosing: relay states, runtimes, remaining liters and remaining percentage.
- Maintenance: Last Backwash, system time and time deviation.

- Faults: place all fault binary sensors in a separate section.

10. Updates and troubleshooting

10.1 No measured values

- Verify that the USR-K5 gateway target is the Home Assistant IP address and TCP port 47524.
- Check whether a previous Node-RED flow is still listening on TCP port 47524.
- Check whether a firewall blocks TCP port 47524.
- Restart Home Assistant and the gateway if required.

10.2 Entity is unavailable or unknown

- An unavailable container value can be expected while no pump flow rate is configured and dosing runtime has already been recorded.
- Last Backwash remains unknown until at least one complete backwash cycle has been detected.
- Orphaned entities may remain in the Home Assistant Entity Registry after updates. Before deleting them, verify whether a new working semantic entity exists.

10.3 Cloud forwarding causes problems

Cloud forwarding is optional. Disable it first if problems occur. Local processing should continue to work. Then review the hostname, port, DNS resolution and firewall rules.

10.4 Repository and support

Project page: https://github.com/JS-DE-Tech/hacs-aseko-asin-aqua-home-clf

Use the GitHub repository issue tracker to report errors and enhancement requests.

Summary

For initial commissioning, four steps are essential: install the HACS integration, disable any previous Node-RED TCP listener, point the USR-K5 gateway to the Home Assistant IP address on port 47524, and initialize the containers with the corresponding replacement buttons.